MULTI QUEUE Scheduling

```
#include <stdio.h>
#define MAX 20

struct Process {
   int pid, at, bt, ct, tat, wt, type;
};

int main() {
   struct Process p[MAX], sq[MAX], uq[MAX];
   int n, i, sq_count = 0, uq_count = 0, time = 0;
   printf("USN:1WA24CS198");
   printf("\nEnter number of processes: ");
   scanf("%d", &n);

   for (i = 0; i < n; i++) {
      printf("\nProcess %d\n", i + 1);
      p[i].pid = i + 1;

      printf("Arrival Time: ");
      scanf("%d", &p[i].at);

      printf("Burst Time: ");
      scanf("%d", &p[i].bt);

      printf("Type (1=System, 2=User): ");
      scanf("%d", &p[i].type);

      if (p[i].type == 1)
         sq[sq_count++] = p[i];
      else
         uq[uq_count++] = p[i];
   }

   // System Queue Execution
   for (i = 0; i < sq_count; i++) {
      if (time < sq[i].at)
         time = sq[i].at;

      time += sq[i].bt;
      sq[i].ct = time;
   }

   // User Queue Execution
   for (i = 0; i < uq_count; i++) {
      if (time < uq[i].at)
         time = uq[i].at;

      time += uq[i].bt;
      uq[i].ct = time;
   }

   float total_tat = 0, total_wt = 0;
```

```c
    printf("\n---System Queue---\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for (i = 0; i < sq_count; i++) {
        sq[i].tat = sq[i].ct - sq[i].at;
        sq[i].wt = sq[i].tat - sq[i].bt;

        total_tat += sq[i].tat;
        total_wt += sq[i].wt;

        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
               sq[i].pid, sq[i].at, sq[i].bt,
               sq[i].ct, sq[i].tat, sq[i].wt);
    }

    printf("\n---User Queue---\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for (i = 0; i < uq_count; i++) {
        uq[i].tat = uq[i].ct - uq[i].at;
        uq[i].wt = uq[i].tat - uq[i].bt;

        total_tat += uq[i].tat;
        total_wt += uq[i].wt;

        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
               uq[i].pid, uq[i].at, uq[i].bt,
               uq[i].ct, uq[i].tat, uq[i].wt);
    }

    printf("\nAverage Turnaround Time = %.2f", total_tat / n);
    printf("\nAverage Waiting Time = %.2f\n", total_wt / n);


    return 0;
}
```

Output

```
USN:1WA24CS198
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 4
Type (1=System, 2=User): 1

Process 2
Arrival Time: 0
Burst Time: 3
Type (1=System, 2=User): 1

Process 3
Arrival Time: 0
Burst Time: 8
Type (1=System, 2=User): 2

Process 4
Arrival Time: 10
Burst Time: 5
Type (1=System, 2=User): 1

---System Queue---
PID AT  BT  CT  TAT WT
1   0   4   4   4   0
2   0   3   7   7   4
4   10  5   15  5   0

---User Queue---
PID AT  BT  CT  TAT WT
3   0   8   23  23  15

Average Turnaround Time = 9.75
Average Waiting Time = 4.75
```